

1. A shepherd boy gets bored tending to his flock to have some fun he cries out "wolf" even though, no is wolf in sight. The villagers run to protect flock, but then they get really not when they realize the boy was playing a joke identify TP, FP, FN

Actual \ Predicted	wolf = yes	wolf = NO
	TP prediction is correct wolf came	FN prediction went wrong wolf came
wolf = yes		
wolf = NO	FP prediction went wrong wolf didn't come	TN prediction correct wolf didn't come

2. Assume the are 100 images, 30 of them depict a cat the rest do not. A ML model predicts the occurrence of a cat in 25 of 30 cat images it also no cat images identify TP, FP, FN, TN

Actual \ Predicted	Cat=yes	Cat=NO
	25	5
Cat=yes	25	5
Cat=NO	20	50
	20	50
	45	55
		100

i) precision

$$\frac{TP}{TP + FP} = \frac{25}{45} = 0.5$$

$$\text{precision} = 0.555 = 55.5\%$$

ii) recall

$$\frac{TP}{TP + FN} = \frac{25}{25 + 5} = \frac{25}{30} = 0.833$$

$$\text{Recall} = 83.33\%$$

iii) F2 Score

$$\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times 55.5 \times 83.33}{55.55 + 83.33}$$

$$= 66.69\%$$

An email service provider wants to classify mails as spam or not spam using features

- * contains "offer" (yes/no)
- * contains "free" (yes/no)
- * Email "length" (short/long)

i) prior probabilities

$$\text{spam} = 3 \quad \text{total} = 6$$

$$\text{not spam} = 3$$

$$P(\text{spam}) = \frac{3}{6} = 0.5$$

$$P(\text{not spam}) = \frac{3}{6} = 0.5$$

ii) conditional probability

spam

$$P(\text{offer} = \text{Yes} \mid \text{spam}) = 2/3$$

$$P(\text{offer} = \text{No} \mid \text{spam}) = 1/3$$

$$P(\text{free} = \text{Yes} \mid \text{spam}) = 3/3 = 1$$

$$P(\text{free} = \text{No} \mid \text{spam}) = 0/3 = 0$$

$$P(\text{length} = \text{short} \mid \text{spam}) = 2/3$$

$$P(\text{length} = \text{long} \mid \text{spam}) = 1/3$$

Not spam

$$P(\text{offer} = \text{Yes} \mid \text{not spam}) = 1/3$$

$$P(\text{offer} = \text{No} \mid \text{not spam}) = 2/3$$

$$P(\text{free} = \text{Yes} \mid \text{not spam}) = \frac{0}{3} = 0$$

$$P(\text{free} = \text{No} \mid \text{not spam}) = \frac{3}{3} = 1$$

$$P(\text{length} = \text{Yes} \mid \text{not spam}) = \frac{1}{3}$$

$$P(\text{length} = \text{No} \mid \text{not spam}) = \frac{2}{3}$$

iii) Naive Bayes:

Spam:

$$P(O = \text{Yes} \mid \text{Free} = \text{No}, \text{length} = \text{short})$$

$$= \frac{1}{2} \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$= 0$$

Not Spam

$$P(\text{not Spam} \mid \text{offer} = \text{yes}, \text{Free} = \text{No}, \text{length} = \text{short})$$

$$= \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$= \frac{1}{18}$$



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